

- 2 (a) resultant moment = zero / sum of clockwise moments = sum of anticlockwise moments
 resultant force = 0
- B1
 B1 [2]
- (b) shape and orientation correct and forces labelled and arrows correct
 angles correct / labelled
- M1
 A1 [2]
- (c) (i) $T \cos 18^\circ = W$
 $T = 520 / \cos 18^\circ = 547 \text{ N}$ Scale diagram:
 $\pm 20 \text{ N}$
- C1
 A1 [2]
- (ii) $R = T \sin 18^\circ$
 $= 169 \text{ N}$ $\pm 20 \text{ N}$
- A1 [1]
- (d) θ is larger hence $\cos \theta$ is smaller, $T = W / \cos \theta$
 hence T is larger
- M1
 A0 [1]

- 3 (a) weight = $m \times g$